

Community Detection on Signed Graphs: A Bayesian MCMC Coupling Approach



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Outline

1. Motivation & the Bayesian framework
2. The Sankararaman & Baccelli model and weak recovery for community detection
3. MCMC Dynamics & Performance Bounds
4. Open Questions & Conclusion

01

Motivation & the Bayesian framework



Haplotype Assembly

1. Biological Setup: Two Homologous Chromosomes (Haplotypes)

Humans are diploid: they have two copies of each chromosome (homologous chromosomes).

Haplotype 1
(Truth $\Sigma = +$)

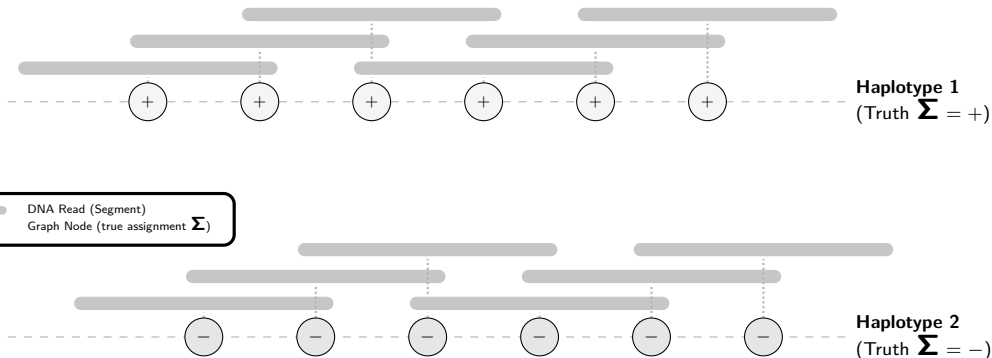
Haplotype 2
(Truth $\Sigma = -$)



Haplotype Assembly

2. High-Throughput Sequencing: DNA Reads

Sequencing generates short, overlapping fragments called **reads** from both chromosomes.



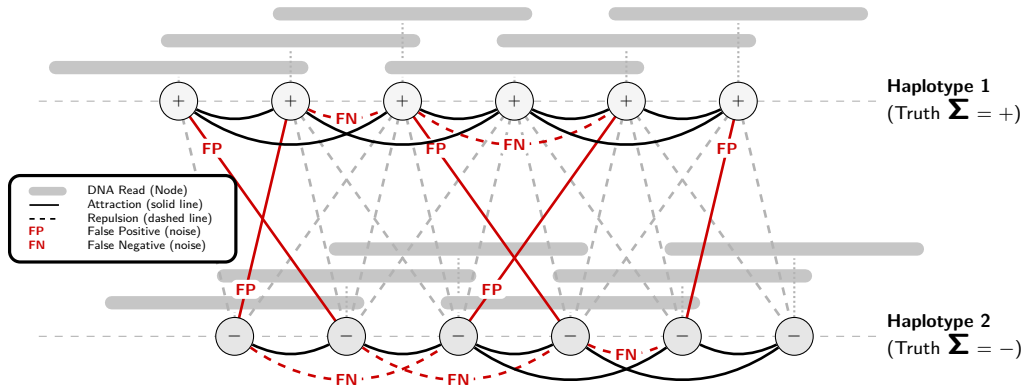


Haplotype Assembly

3. Modeling as a Signed Graph

Nodes = reads Edges = overlaps

Goal: Reconstruct the true haplotype assignments Σ from this observed signed graph.





Motivation: Signed Community Detection

Community Detection on Signed Graphs ($K = 2$)

- ▶ Partition nodes into $K = 2$ communities: $\Sigma_i \in \{+1, -1\}$.
- ▶ Links represent **attractions** (solid) and **repulsions** (dashed).
- ▶ Beyond the classical SBM: a rich framework coupling spatial geometry and community structure!

The Geometric Challenge

- ▶ Spatial embedding (e.g., positions on $\mathbb{Z}$ or $\mathbb{R}^d$).
- ▶ Edge probabilities depend on distance $r = \|x_i - x_j\|$:

$$\mathbb{P}(\text{attractive edge} \mid \text{same community}) = f_{\text{in}}(r)$$

$$\mathbb{P}(\text{attractive edge} \mid \text{different communities}) = f_{\text{out}}(r)$$

- ▶ **Main interest of the model... but also core challenge:** capturing this geometry.

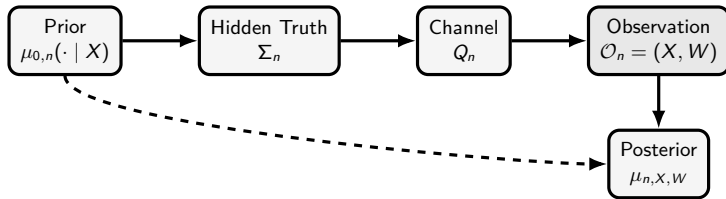


The General Bayesian Model

Definition

1. **Nodes, Edges, Communities:** Nodes $V_n = \{1, \dots, n\}$, edges E_n , communities $\mathcal{L}_K = \{0, \dots, K-1\}$, and set of configurations $\sigma \in \mathcal{C}_n = (\mathcal{L}_K)^{V_n}$.
2. **Features:** $X_n \in \mathcal{X}_n$ according to distribution ν_n .
3. **Prior on configurations:** $\mu_{0,n}(\mathrm{d}\sigma \mid x)$ over $\mathcal{C}_n$.
4. **(Noisy) channel:** $Q_n(\mathrm{d}w \mid x, \sigma) = \prod_{e \in E_n} Q_{e,n}(\mathrm{d}w_e \mid x_i, x_j, \sigma_i, \sigma_j)$.

The joint law is $\mathbb{P}_n(\mathrm{d}x, \mathrm{d}\sigma, \mathrm{d}w) = \nu_n(\mathrm{d}x) \mu_{0,n}(\mathrm{d}\sigma \mid x) Q_n(\mathrm{d}w \mid x, \sigma)$.





Advantages of the Bayesian Framework

Rigorous Definitions

- ▶ **Algorithm** τ_n : MARKOV kernel mapping observations (X, W) to configurations.
- ▶ **Random Guess**: An algorithm agnostic to observations (relying only on the prior μ_0).
- ▶ **Weak / Partial Recovery**: Beating the best random guess asymptotically.

Posterior Resampling

The hidden truth Σ can be replaced by a replica $\Sigma^{(1)}$ drawn from the posterior distribution $\mu_{n,X,W}$ (NISHIMORI identity).

Measure of Performance: Overlap

The overlap quotients the HAMMING similarity by label permutations $\mathfrak{S}_K$:

$$\text{ov}_n(\sigma, \sigma') = \max_{\pi \in \mathfrak{S}_K} \frac{1}{n} \sum_{i=1}^n \mathbb{I}_{\{\sigma'_i = \pi(\sigma_i)\}}$$

- ▶ Uniform prior: best random guess score is $1/K$.
- ▶ Weak recovery: $\text{overlap} \geq 1/K + \eta$.



Posterior Resampling (Nishimori Identity)

Core Idea: Replacing Truth by a Replica

Comparing τ_n with hidden truth $\Sigma_n \iff$ Comparing τ_n with posterior replica $\Sigma_n^{(1)} \sim \mu_{n, X_n, W_n}$

Theorem (Nishimori Identity [1])

For any algorithm τ_n and replica $\Sigma_n^{(1)} \sim \mu_{n, X_n, W_n}$:

$$\mathbb{E} [\Psi_n(X_n, W_n, \Sigma_n, \tau_n)] = \mathbb{E} [\Psi_n(X_n, W_n, \Sigma_n^{(1)}, \tau_n)]$$

Corollary (Overlap Identity)

For any threshold $s \in [0, 1]$:

$$\mathbb{P} [\text{ov}_n(\Sigma_n, \tau_n) \geq s] = \mathbb{P} [\text{ov}_n(\Sigma_n^{(1)}, \tau_n) \geq s]$$

02

The Sankararaman & Baccelli model and weak recovery for community detection





The Sankararaman–Baccelli Model (GSBM)

Model Definition [2]

- **Nodes:** Realization X of a homogeneous POISSON point process in an n -volume of $\mathbb{R}^d$ [3].
- **Latent Communities:** Uniform assignments $\Sigma_i \in \{+1, -1\}$ ($K = 2$).
- **Edges:** Connection $e = \{i, j\} \in H$ occurs with probability:

$$\mathbb{P}(e \in H \mid X, \Sigma) = \begin{cases} f_{\text{in}}(r_e) & \text{if } \Sigma_i = \Sigma_j \text{ (attraction),} \\ f_{\text{out}}(r_e) & \text{if } \Sigma_i \neq \Sigma_j \text{ (repulsion),} \end{cases}$$

where $r_e = \|X_i - X_j\|$ and $f_{\text{in}}(r) \geq f_{\text{out}}(r)$.

Goal: Weak Recovery

- Find an algorithm τ_n (independent of Σ_n) achieving:

$$\lim_{n \rightarrow \infty} \mathbb{E}[\text{ov}_n(\Sigma_n, \tau_n)] \geq \frac{1}{2} + \eta \quad (\eta > 0), \quad \text{where } \text{ov}_n(\Sigma_n, \tau_n) = \max_{\pi \in \mathfrak{S}_2} \frac{1}{n} \sum_{i=1}^n \mathbb{I}_{\{\tau_{n,i} = \pi(\Sigma_{n,i})\}}.$$

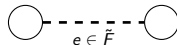
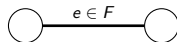


Gibbs Model on Signed Graphs

Attractive and Repulsive Edges

Partition potential edges $E = F \dot{\cup} \tilde{F}$:

- ▶ F : **Attractive** (ferromagnetic), weight $W_e > 0$
- ▶ $\tilde{F}$: **Repulsive** (antiferromagnetic), weight $W_e < 0$



Signed Gibbs Measure [4]

Configuration probability $\mu(\sigma) = \frac{1}{Z} e^{-U(\sigma)}$, where energy $U(\sigma)$ penalizes unsatisfied edges:

$$U(\sigma) = \sum_{e=\{i,j\} \in F} W_e \mathbb{I}_{\{\sigma_i \neq \sigma_j\}} + \sum_{e=\{i,j\} \in \tilde{F}} |W_e| \mathbb{I}_{\{\sigma_i = \sigma_j\}}$$

Weights $|W_e|$ dictate constraint strengths.



From GSBM to the Gibbs Posterior

Conditioning on the Point Cloud

- ▶ Positions X are fixed $\implies$ all distances $r_e = \|X_i - X_j\|$ are known.
- ▶ Under a uniform prior, the posterior given the observed graph H satisfies:

$$\mathbb{P}(\Sigma = \sigma \mid H, X) \propto \mathbb{P}(H \mid X, \sigma)$$

Log-Likelihood as Signed Gibbs Energy

- ▶ Setting edge weights $W_e(H)$:

$$W_e(H) = \begin{cases} \log \left(\frac{f_{\text{in}}(r_e)}{f_{\text{out}}(r_e)} \right) \geq 0 & \text{if } e \in H \text{ (attractive),} \\ -\log \left(\frac{1 - f_{\text{out}}(r_e)}{1 - f_{\text{in}}(r_e)} \right) \leq 0 & \text{if } e \notin H \text{ (repulsive).} \end{cases}$$

- ▶ ... recovers the GIBBS posterior: $\mathbb{P}(H \mid X, \sigma) \propto e^{-U(\sigma)}$.



From GSBM to the Gibbs Posterior

Conditioning on the Point Cloud

- Positions X are fixed $\implies$ all distances $r_e = \|X_i - X_j\|$ are known.
- Under a uniform prior, the posterior given the observed graph H satisfies:

$$\mathbb{P}(\Sigma = \sigma \mid H, X) = \mu(\sigma) \propto \mathbb{P}(H \mid X, \sigma)$$

Log-Likelihood as Signed Gibbs Energy

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The Percolation Obstruction

Necessary Condition for Weak Recovery [2]

- ▶ Global information flow requires the graph to **percolate** [5] (possess a giant component).

Theorem (Sankararaman & Baccelli [2])

Independent bond percolation with edge probability

$$p_e = f_{\text{in}}(r_e) - f_{\text{out}}(r_e)$$

*yielding a giant component is a **necessary condition** for weak recovery.*

- ▶ **Our goal:** Retrieve and extend this result using unified **Bayesian MCMC coupling**.

03

MCMC Dynamics & Performance Bounds

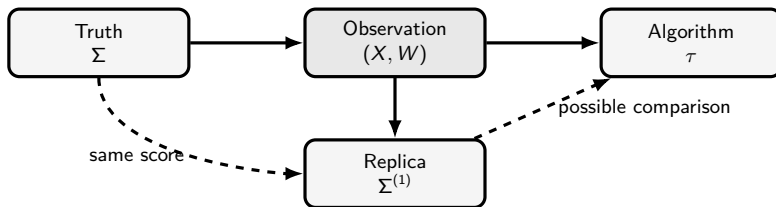




Invariant Coupling

Markov Chain Dynamics as a Source of Coupling

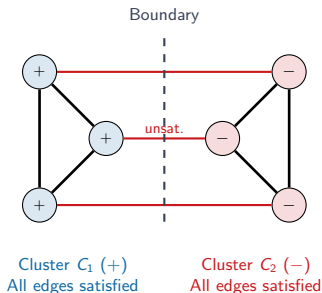
- ▶ We construct the posterior replica $\Sigma^{(1)}$ starting from the hidden truth Σ .
- ▶ Running **exactly one step** of a posterior-invariant MARKOV kernel M initialized at Σ yields a valid replica $\Sigma^{(1)} \sim M(\Sigma, \cdot)$.



Single-Spin Dynamics & Geometric Bottleneck

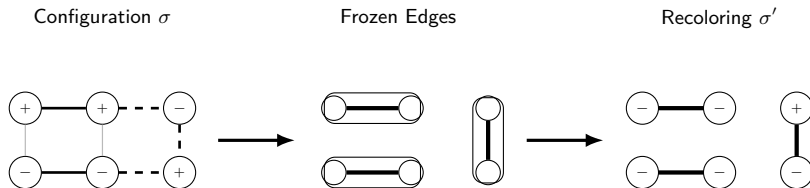
Single-Spin Update Limitations

- ▶ **Geometry makes updates slow:** Single-spin dynamics (GLAUBER / METROPOLIS-HASTINGS [6]) update one node at a time.
- ▶ **Contradictory Clusters:** Two internally valid (highly satisfied) but mutually contradictory clusters are locked.
- ▶ **High Energy Barrier:** Flipping a single boundary node violates internal edges, creating a local energy barrier.
- ▶ **Conclusion:** Need for geometry $\implies$ **Cluster Dynamics** (updating whole regions).





One Step of Signed Swendsen–Wang [7, 8]



- Edges in bold represent satisfied interactions that are randomly frozen.
- Clusters are then flipped independently.



Swendsen–Wang Signed Cluster Dynamics

Swendsen–Wang (SW) Signed Cluster Dynamics [7, 8]

- ▶ Update whole clusters of vertices simultaneously to speed up mixing.
- ▶ **Freezing Step:** For each edge $e = \{i, j\}$:
 - If e is satisfied ($W_e > 0$ and $\sigma_i = \sigma_j$, or $W_e < 0$ and $\sigma_i \neq \sigma_j$), freeze it with probability:

$$p_e^{\text{frozen}} = 1 - e^{-|W_e|}.$$

- If e is unsatisfied, do not freeze it ($p_e^{\text{frozen}} = 0$).
- ▶ **Recoloring Step:** Flip each frozen cluster independently with probability $1/2$.



High-Order Triangular Dynamics

The Frustration Obstruction

- ▶ **Frustration:** $\nexists \sigma$ satisfying all edges (conflicting constraints).
- ▶ **Excessive Freezing:** In frustrated configurations, standard SW satisfies and freezes too many edges independently, blocking/freezing the MCMC dynamics.
- ▶ **Higher-Order Correlation:** Group interactions into **triangles** to correlate freezing and **freeze** as little as possible.

Triangular Cluster Dynamics [9]

- ▶ **Energy Decomposition:** Distribute edge energy U over triangles: $U(\sigma) = \sum_{t \in \mathcal{T}} U_t(\sigma)$.
- ▶ **Freezing Mechanism:** Apply a joint freezing mechanism on the triangles (rather than independent edges) to keep cluster sizes active.



Triangular Dynamics Freezing Rules

$\triangle$ Attractive Triangle (Isotropic)					
Configuration	$\emptyset$	$\{ab\}$	$\{ac\}$	$\{bc\}$	$\{ab, ac, bc\}$
$(+, +, +)$	b_u	0	0	0	a_u
Other	1	0	0	0	0

$\triangleleft \triangle$ Repulsive Triangle (Isotropic)					
Configuration	$\emptyset$	$\{ab\}$	$\{ac\}$	$\{bc\}$	$\{ab, ac, bc\}$
$(+, +, +)$	1	0	0	0	0
$(+, +, -)$	b_u	0	$a_u/2$	$a_u/2$	0
$(+, -, +)$	b_u	$a_u/2$	0	$a_u/2$	0
$(+, -, -)$	b_u	$a_u/2$	$a_u/2$	0	0

Table: Freezing rules for triangular dynamics ($a_u = 1 - e^{-2u}$, $b_u = e^{-2u}$), where u is the energy transferred by each edge.

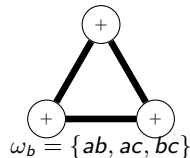
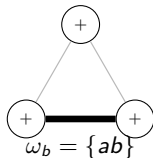
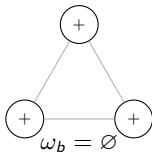


General Swendsen–Wang Cluster Dynamics: Framework

General Framework [10]

Suppose the Hamiltonian decomposes as: $U(\sigma) = U_0(\sigma) + \sum_{b \in \mathcal{B}} U_b(\sigma)$, where each $b \in \mathcal{B}$ is a **bond** (e.g., edge, triangle, clique).

1. **Freezing Step:** For each bond $b \in \mathcal{B}$, freeze a sub-bond $\omega_b \subseteq b$ with probability $P_b^\sigma(\omega_b)$.
2. **Recoloring Step:** Recolor each frozen cluster independently according to a distribution preserving the prior $\mu_0(\sigma) \propto e^{-U_0(\sigma)}$.





General Swendsen–Wang Cluster Dynamics: Invariance

Definition (Local Reversibility)

For two configurations σ, σ' and a bond b , we define the compatibility probability:

$$P_b^{\sigma \rightarrow \sigma'}(\omega_b) = \begin{cases} P_b^\sigma(\omega_b), & \text{if } P_b^{\sigma'}(\omega_b) > 0, \\ 0, & \text{otherwise.} \end{cases}$$

The freezing rule is **locally reversible** if for all $b, \omega_b, \sigma, \sigma'$:

$$P_b^{\sigma \rightarrow \sigma'}(\omega_b) = e^{-(U_b(\sigma') - U_b(\sigma))} P_b^{\sigma' \rightarrow \sigma}(\omega_b)$$

Theorem (Gibbs Invariance [10])

If freezing choices are independent, locally reversible, and $P_b^\sigma(\emptyset) > 0$ for all b, σ :

1. The global cluster dynamics is reversible with respect to $\mu(\sigma) \propto e^{-U(\sigma)}$.
2. The condition $P_b^\sigma(\emptyset) > 0$ ensures the **irreducibility** of the MARKOV chain.

Percolation as a Necessary Condition for Recovery

Theorem (Percolation Bound on Overlap [10])

Under the uniform prior μ_0 , if frozen clusters are recolored independently and uniformly, then for the one-step posterior replica $\Sigma_n^{(1)}$, any algorithm τ_n , and any $\eta > 0$:

$$\lim_{n \rightarrow \infty} \mathbb{P} \left[\text{ov}_n(\Sigma_n^{(1)}, \tau_n) \geq \frac{1}{K} + \frac{K-1}{K}(\theta^{\max} + \eta) \right] = 0$$

where $\theta^{\max}$ is the asymptotic fraction of nodes in giant frozen components (which bounds the maximum recoverable fraction of communities).

Corollary (Percolation is Necessary)

*If there is no percolation ($\theta^{\max} = 0$), the maximum overlap is bounded by $1/K$ (random guess): weak recovery is impossible. Hence, **percolation is a necessary condition for weak recovery**.*

Key Intuition

Small clusters are recolored independently, washing out their contribution. Only giant (macroscopic) components can carry global community information.



Coupling & the Sankararaman–Baccelli Bound

One Step of Classical Edge-by-Edge Swendsen–Wang

Let's run one step of standard SWENDSEN–WANG (edges) initialized at the hidden truth Σ . For any edge e , the probability that it is frozen is:

► **Case 1** ($\Sigma_i = \Sigma_j$):

$$\mathbb{P}(\text{frozen}) = f_{\text{in}}(r_e) \left(1 - e^{-|W_e|}\right) = f_{\text{in}}(r_e) \left(1 - \frac{f_{\text{out}}(r_e)}{f_{\text{in}}(r_e)}\right) = f_{\text{in}}(r_e) - f_{\text{out}}(r_e)$$

► **Case 2** ($\Sigma_i \neq \Sigma_j$):

$$\mathbb{P}(\text{frozen}) = (1 - f_{\text{out}}(r_e)) \left(1 - e^{-|W_e|}\right) = (1 - f_{\text{out}}(r_e)) \left(1 - \frac{1 - f_{\text{in}}(r_e)}{1 - f_{\text{out}}(r_e)}\right) = f_{\text{in}}(r_e) - f_{\text{out}}(r_e)$$

Corollary (Retrieving the S&B Bound [2])

*Under the classical edge SW dynamics, the frozen graph is **exactly** an independent bond percolation with edge probability:*

$$p_e = f_{\text{in}}(r_e) - f_{\text{out}}(r_e).$$



Strict Improvement of the S&B bound (on the triangular grid)

Homogeneous Model ($f_{\text{in}} = p$, $f_{\text{out}} = 1 - p$)

On the triangular grid, we select a family of independent white triangles (like a chessboard).

- **Edge Dynamics (Standard SW):** Frozen graph is independent edge percolation with parameter $p_{\text{edge}} = 2p - 1$. Critical threshold:

$$p_c^{\text{edge}} = \frac{1}{2} + \sin\left(\frac{\pi}{18}\right) \approx 0.673 \dots$$

- **Triangular Dynamics:** Frozen graph yields correlated triangle percolation. Critical threshold:

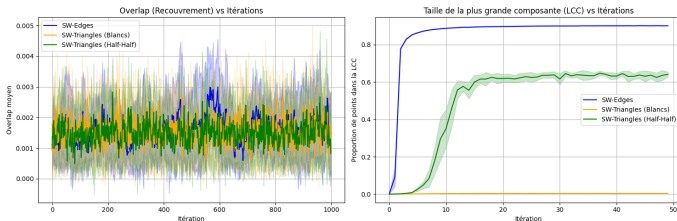
$$p_c^{\triangle} = \frac{7}{4} - \frac{\sqrt{17}}{4} \approx 0.7192 \dots$$

Corollary (Strictly Better Impossibility Bound [10])

Since $p_c^{\text{edge}} < p_c^{\triangle}$, for any $p \in [0.673 \dots, 0.7192 \dots)$, the standard S&B bound cannot conclude (edge dynamics percolate), whereas triangular dynamics do not, proving that weak recovery is impossible.



Empirical Results: Subcritical vs. Supercritical ($p = 0.70$)



Phase Space

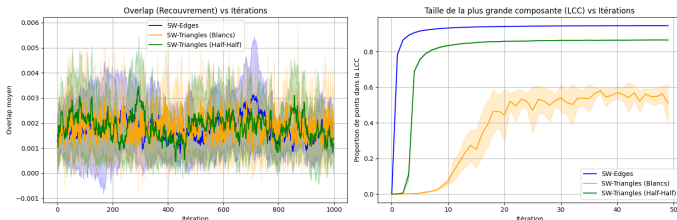
$(p_c^{\text{edge}} \approx 0.673 < 0.70 < p_c^{\Delta} \approx 0.7192)$

- ▶ **Edges (standard SW):** supercritical ($\theta^{\max} > 0$), largest component size (LCC) reaches ≈ 0.9 (right plot).
- ▶ **Triangles / Split weight:** subcritical ($\theta^{\max} = 0$), LCC remains near 0.

Observed Behavior

- ▶ Only standard edge SW achieves giant components.
- ▶ Overlap remains at the baseline noise level (around 0.002) for triangles and split weights due to lack of percolation (left plot).

Empirical Results: Onset of Percolation ($p = 0.72$)



Transition Regime

($p = 0.72 > p_c^\Delta \approx 0.7192$)

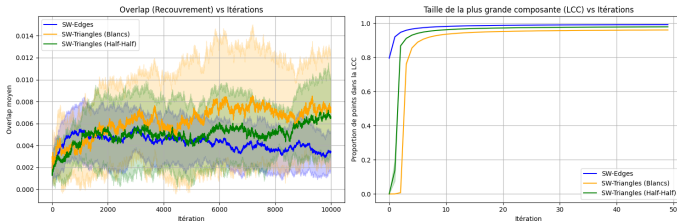
- ▶ All three dynamics are now **supercritical**.
- ▶ LCC grows for all: edges reach ≈ 0.9 , split weights reach ≈ 0.85 , and white triangles reach ≈ 0.55 (right plot).

Observed Behavior

- ▶ Giant components have formed, but the overlap remains low (below 0.003) for all dynamics (left plot).
- ▶ **No weak recovery yet:** the true weak recovery threshold is likely strictly above the percolation threshold (percolation is only a necessary condition).



Empirical Results: Weak Recovery Achieved ($p = 0.80$)



Supercritical Regime ($p = 0.80 \gg p_c^\Delta$)

- Giant components are well-established (right plot), with LCC close to 0.95 for edges and split weights, and 0.90 for white triangles.
- Fluctuations are small and components are stable.

Observed Behavior

- Weak recovery is successfully achieved for all algorithms.
- The overlap climbs significantly above the noise level (left plot), peaking at ≈ 0.010 for white triangles and ≈ 0.006 for split weights.

04

Open Questions & Conclusion





Open Questions

Limits of the One-Replica Coupling

- ▶ Our framework constructs **one** replica $\Sigma_n^{(1)}$ using MCMC cluster dynamics.
- ▶ This only yields an impossibility bound (**necessary condition** for recovery).

Towards a Necessary and Sufficient Condition

- ▶ Achieving a **necessary and sufficient** condition requires a **two-replica** framework.
- ▶ Open: How to define and analyze two-replica couplings under MCMC?

Towards a Practical Community Detection Algorithm

Limits of Direct Gibbs Sampling

- ▶ Direct posterior sampling does not perform well in practice.
- ▶ Likely due to slow mixing of the underlying MARKOV chain.

Estimating the Correlation Matrix

- ▶ Estimating the correlation matrix $\mathbb{P}(\sigma_i = \sigma_j)$ yields much better results.
- ▶ **Future work:** Understand why correlation estimators are more robust.



Distance Adapted to Cluster Dynamics

Hamming Similarity on the Quotient Space

- ▶ Standard overlap is a **HAMMING** similarity quotiented by label permutations:

$$\text{ov}_n(\sigma, \sigma') = \max_{\pi \in \mathfrak{S}_K} \frac{1}{n} \sum_{i=1}^n \mathbb{I}_{\{\sigma'_i = \pi(\sigma_i)\}}.$$

Markov Transition-Adapted Distance

- ▶ Consider distances reflecting transitions of the cluster dynamics.
- ▶ For standard SW, this corresponds to the **symmetric difference** on satisfied edges:

$$d_{\text{sym}}(\sigma, \sigma') = \frac{1}{2} \sum_{e=\{i,j\}} |W_e| \left| \mathbb{I}_{\{\sigma \text{ satisfies } e\}} - \mathbb{I}_{\{\sigma' \text{ satisfies } e\}} \right|$$

- ▶ Establishes links to **spectral graph analysis** (e.g., LAPLACIAN eigenvalues).

Application: 3-SAT Solving via Signed Graphs

The 3-SAT Problem

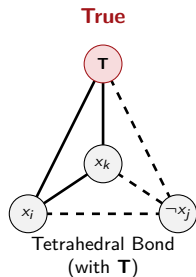
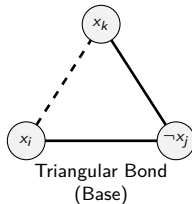
Find a truth assignment satisfying a set of clauses, e.g.,

$$\mathcal{C} = x_i \vee \neg x_j \vee x_k$$

Clause Encoding

We introduce a reference node **T** (True):

- ▶ **Triangle** (literals $x_i, \neg x_j, x_k$): edge signs opposite to the literal product.
- ▶ **Tetrahedron (with T)**: edge signs conforming to the literal signs.





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